Chapter 3 solutions

CH03-E1

1. S_1 is a line through the origin. For $(x, 2x)^T$, $(z, 2z)^T$ and $c \in \mathbb{R}$,

$$(x, 2x)^T + (z, 2z)^T = (x + z, 2x + 2z)^T \in S_1,$$
$$c(x, 2x)^T = (cx, 2cx)^T \in S_1.$$

Also $0 \in S_1$, so it is a subspace.

2. S_2 is not a subspace. Both $(1, 0)^T$ and $(0, 1)^T$ belong to it, but

$$(1, 0)^T + (0, 1)^T = (1, 1)^T \notin S_2.$$

Addition is not closed. Containing zero alone is insufficient.

3. S_3 is not a subspace. It contains $(1, 0)^T$, but multiplication by 2 gives $(2, 0)^T \notin S_3$. It also omits zero. The unit circle fails the linear closure conditions.

CH03-E2

1. Subtract twice row 1 from row 2, and row 1 from row 3.

$$A \to \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

No further pivot-column entries need clearing, so RREF is

$$R = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The rank is 2. Pivot columns are 1 and 3; free columns are 2 and 4.

2. Set $x_2 = s, x_4 = t$. Then

$$x_1 + 2s + t = 0, \qquad x_3 + t = 0.$$

Therefore

$$x = (-2s - t, s, -t, t)^T = s(-2, 1, 0, 0)^T + t(-1, 0, -1, 1)^T.$$

A basis is

$$\mathcal{B}_{N(A)} = \{(-2, 1, 0, 0)^T, (-1, 0, -1, 1)^T\}.$$

3. Direct multiplication gives

$$A(-2, 1, 0, 0)^T = (-2 + 2, -4 + 4, -2 + 2)^T = 0,$$
$$A(-1, 0, -1, 1)^T = (-1 + 1, -2 - 1 + 3, -1 - 1 + 2)^T = 0.$$

If $c_1 s_1 + c_2 s_2 = 0$, coordinate 2 forces $c_1 = 0$ and coordinate 4 forces $c_2 = 0$. Hence the vectors are independent.

Chapter 3 solutions (continued)

CH03-E3

Augment the matrix by b.

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & 1 & 3 & 5 \\ 1 & 2 & 1 & 2 & 3 \end{array}\right].$$

Subtract twice row 1 from row 2, and row 1 from row 3.

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \end{array}\right].$$

Subtract row 2 from row 3. The left block is RREF and the right block is $(2,1,0)^T$. There is no contradictory row, so the system is consistent.

Set $x_2 = x_4 = 0$. Then

$$x_1 = 2, \qquad x_3 = 1$$

giving $x_p = (2,0,1,0)^T$.

The complete solution is

$$x = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \quad s,t \in \mathbb{R}.$$

Since $b \neq 0$, zero is absent. This is a translate of $N(A)$, not a subspace.

CH03-E4

1. Original columns 1 and 3 are the pivot columns.

$$\mathcal{B}_{C(A)} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

They are independent, so they form a basis and $\dim C(A) = 2$.

2. The nonzero RREF rows give

$$\mathcal{B}_{C(A^T)} = \{(1,2,0,1)^T, (0,0,1,1)^T\}.$$

At coordinates 1 and 3, these vectors have patterns $(1,0)$ and $(0,1)$. They are independent, so $\dim C(A^T) = 2$.

3. Solve $A^T y = 0$ with $y = (a,b,c)^T$. The first three equations are below. The fourth, $a + 3b + 2c = 0$, is the sum of the first and third and adds no constraint.

$$a + 2b + c = 0, \quad 2a + 4b + 2c = 0, \quad b + c = 0.$$

Chapter 3 solutions (continued)

The second equation repeats twice the first. The third gives $c = -b$, and the first then gives $a = -b$. Hence

$$N(A^T) = \text{span}\{(-1, 1, -1)^T\},$$

This gives a one-vector basis.

4. Since $m = 3, n = 4, r = 2$,

$$r + (n - r) = 2 + 2 = 4 = n,$$
$$r + (m - r) = 2 + 1 = 3 = m.$$

The first sum pairs row space and nullspace in $\mathbb{R}^4$. The second pairs column space and left nullspace in $\mathbb{R}^3$.

CH03-E5

The trigonometric identity gives

$$\sin^2 x = \frac{1 - \cos(2x)}{2}.$$

Therefore

$$\frac{1}{2}v_1 - \frac{1}{2}v_2 - v_3 = 0$$

for every x. The coefficients $(1/2, -1/2, -1)$ are not all zero, proving dependence.

Since v_3 is generated by v_1, v_2, the span has dimension at most 2. If $a + b\cos(2x) = 0$ for every x, setting $x = 0$ gives $a + b = 0$, and setting $x = \pi/4$ gives $a = 0$, hence $b = 0$. Thus v_1, v_2 are independent and form a basis of a two-dimensional span.

Chapter 3 solutions (continued)

CH03-E6

1. With $m = 3, n = 5, r = 2$, the dimensions are as follows.
   - $C(A) \subseteq \mathbb{R}^3$, dimension 2;
   - $C(A^T) \subseteq \mathbb{R}^5$, dimension 2;
   - $N(A) \subseteq \mathbb{R}^5$, dimension $5 - 2 = 3$;
   - $N(A^T) \subseteq \mathbb{R}^3$, dimension $3 - 2 = 1$.
2. Every consistent solution set is $x_p + N(A)$. Since $\dim N(A) = 3$, it has three free parameters.
3. Yes. Rank $2 < 3$ makes $C(A)$ a proper plane in $\mathbb{R}^3$. Every b outside it is inconsistent. Nonzero homogeneous solutions do not imply that all right-hand sides are attainable.
4. No. If $\dim N(A) = 0$, rank-nullity would give
$$r = 5 - \dim N(A) = 5.$$

This exceeds the number of rows, 3, and contradicts the given rank 2. In fact nullity equals 3.

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CH03-E7

Let $c = (c_1, c_2, c_3)^T$. Independence of the given basis means that
$$c_1 w_1 + c_2 w_2 + c_3 w_3 = 0$$
is equivalent to $Bc = 0$.

1. B has pivots in columns 1 and 2 and rank 2. Thus $Bc = 0$ has a nonzero solution, so the three w_j are dependent.
2. The coordinate isomorphism preserves span dimension, giving dimension $\operatorname{rank} B = 2$. Here $w_1 = u_1, w_2 = u_2$. Their coordinate vectors are the first two standard basis vectors, so they form a basis of this span.
3. The third column of B is the sum of its first two columns.
$$\operatorname{col}_3(B) = \operatorname{col}_1(B) + \operatorname{col}_2(B).$$

Hence $B(1, 1, -1)^T = 0$, giving
$$w_1 + w_2 - w_3 = 0.$$
Substituting $w_3 = w_1 + w_2$ verifies the relation directly. The coefficients $(1, 1, -1)$ are nonzero as a vector.

Chapter 3 solutions (continued)

CH03-E8

1. Using the four basis vectors from the chapter,

$$u_1 = (1,2,0,1)^T, \quad u_2 = (0,0,1,1)^T,$$
$$s_1 = (-2,1,0,0)^T, \quad s_2 = (-1,0,-1,1)^T.$$

coordinate addition of $u_1 + u_2 + s_1 + s_2$ gives

$$x_{\text{row}} = u_1 + u_2 = (1,2,1,2)^T,$$
$$x_{\text{null}} = s_1 + s_2 = (-3,1,-1,1)^T.$$

so $x = (-2,3,0,3)^T$.

2. x_{row} is the sum of the two nonzero RREF rows, hence belongs to $C(A^T)$. Also

$$As_1 = As_2 = 0,$$

so $x_{\text{null}} \in N(A)$. Their sum is x.

3. Suppose

$$\alpha u_1 + \beta u_2 + \gamma s_1 + \delta s_2 = 0,$$

Comparing coordinates gives

$$\gamma = -2\alpha, \quad \delta = \beta, \quad 5\alpha - \beta = 0, \quad \alpha + 2\beta = 0.$$

Thus $\beta = 5\alpha$; substitution in the fourth equation gives $11\alpha = 0$, so all coefficients vanish. These four independent vectors form a basis of $\mathbb{R}^4$, giving a decomposition for every x. The difference of two decompositions is a zero combination of this basis, forcing every coefficient difference to vanish and proving uniqueness.

Chapter 3 solutions (continued)

Verification code appendix: four_subspaces.py

```
"""Verify rank and basis dimensions for the chapter-three example."""

from fractions import Fraction


A = [
    [1, 2, 0, 1],
    [2, 4, 1, 3],
    [1, 2, 1, 2],
]

NULL_BASIS = [
    [-2, 1, 0, 0],
    [-1, 0, -1, 1],
]

LEFT_NULL_BASIS = [
    [-1, 1, -1],
]


def dot(left, right):
    return sum(x * y for x, y in zip(left, right))


def matvec(matrix, vector):
    return [dot(row, vector) for row in matrix]
```

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Chapter 3 solutions (continued)

```
first_two_columns = [[row[0], row[1]] for row in A]
assert all(pair[1] == 2 * pair[0] for pair in first_two_columns)
assert [row[2] for row in A] == [0, 1, 1]
assert all(matvec(A, vector) == [0, 0, 0] for vector in NULL_BASIS)
assert all(
    dot([row[column] for row in A], LEFT_NULL_BASIS[0]) == 0
    for column in range(len(A[0]))
)

column_basis = [
    [row[0] for row in A],
    [row[2] for row in A],
]
assert [Fraction(x) for x in column_basis[0]] == [1, 2, 1]
assert [Fraction(x) for x in column_basis[1]] == [0, 1, 1]

rank = 2
assert rank + len(NULL_BASIS) == len(A[0]) == 4
assert rank + len(LEFT_NULL_BASIS) == len(A) == 3

print("rank=2; dim C(A)=dim C(A^T)=2")
print("dim N(A)=2; dim N(A^T)=1")
```

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References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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